

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To implement a 2-bit asynchronous Counter using D Flip-Flops and observe SA0 and SA1 Faults.	
Experiment No: 10	Date: 03-04-2026	Enrolment No: 92301733056

AIM: To implement a 2-bit asynchronous counter using D Flip-Flops and observe SA0 and SA1 Faults.

IDE: EDA Playground

THEORY:

A 2-bit asynchronous counter using D Flip-Flops is a sequential digital circuit that counts binary numbers from 00 to 11 based on clock pulses. In an asynchronous or ripple counter, only the first flip-flop is driven by the external clock, while the next flip-flop is triggered by the output of the previous one, causing a ripple effect. A D Flip-Flop follows the characteristic equation $Q_{next} = D$. To make it function as a toggle flip-flop for counting, the input is connected as $D = \bar{Q}$, so the output changes state with each clock pulse. In this design, two D Flip-Flops are used: the first flip-flop (LSB) toggles with every clock pulse, and its output is given as the clock to the second flip-flop (MSB). This results in the counting sequence $00 \rightarrow 01 \rightarrow 10 \rightarrow 11$.

DESIGN AND IMPLEMENTATION:

The design and implementation of a 2-bit asynchronous counter using D Flip-Flops involves connecting two D Flip-Flops in cascade to achieve binary counting. Each flip-flop is configured to operate in toggle mode by connecting its complemented output ($\bar{Q}$) back to its D input, ensuring that the output changes state with every triggering clock pulse. The external clock signal is applied to the first flip-flop (LSB), while the output of this flip-flop (Q0) is used as the clock input for the second flip-flop (MSB), creating the ripple effect characteristic of asynchronous counters. The outputs Q0 and Q1 represent the least and most significant bits, respectively, and LEDs can be connected to these outputs to observe the counting sequence. During implementation, the circuit is first tested under normal conditions to verify the sequence $00 \rightarrow 01 \rightarrow 10 \rightarrow 11$. To analyze fault behavior, stuck-at-0 (SA0) and stuck-at-1 (SA1) faults are introduced by forcing selected nodes to logic 0 or logic 1, and the resulting output variations are observed and compared with the expected sequence to study the impact of faults on circuit performance.

VERILOG CODE:

2-bit asynchronous counter

A. Verilog Code:

```
// Code your design here
module async_counter(
    input clk,
    input reset,
    output [1:0] q
);

    wire q0,q1;

    //1st flipflop
    d_ff ff0(
        .clk(clk),
```

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```

.reset(reset),
.d(~q0),
.q(q0)
);

//2nd flipflop
d_ff ff1(
.clk(clk),
.reset(reset),
.d(~q1),
.q(q1)
);

assign q={q1,q0};
endmodule

//d flipflop module
module d_ff(
input clk, reset, d,
output reg q
);

always @(posedge clk or posedge reset)
begin
if(reset)
q<= 0;
else
q<= d;
end
endmodule

```

B. Testbench:

```

// Code your testbench here
// or browse Examples
`timescale 1ns/1ps
module tb_async_counter;
reg clk;
reg reset;
wire [1:0] q;

// instantiate the DUT
async_counter uut(
.clk(clk),
.reset(reset),
.q(q1)
);

//clock generation
always #5 clk = ~clk;

```

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```

initial
begin
  clk = 0;
  reset = 1;

  #10;
  reset = 0; // Apply reset

  #100;

  $finish;
end

// Dump waveforms
initial begin
  $dumpfile("dump.vcd");
  $dumpvars(0,tb_async_counter);
end

//Monitor values
initial begin
  $monitor("time=%0t | reset = %b | count = %b", $time,reset,q);
end
endmodule

```

Simulation result:



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2-bit asynchronous counter with Fault Injection & Detection

A. Verilog Code:

```
// Code your design here
module async_counter_fault(
    input clk,
    input reset,
    input sa0,
    input sa1,
    output [1:0] q,
    output [1:0] q_fault,
    output error
);

    wire q0,q1;

    // Normal Counter
    d_ff ff0(.clk(clk), .reset(reset), .d(~q0), .q(q0));
    d_ff ff1(.clk(clk), .reset(reset), .d(~q1), .q(q1));

    assign q = {q1,q0};

    // Fault Injection Logic
    assign q_fault[0] = sa0?1'b0:(sa1?1'b1:q0);
    assign q_fault[1] = sa0?1'b0:(sa1?1'b1:q1);

    // Error Detection
    assign error = (q^q_fault);
endmodule

// D flipflop module
module d_ff(
    input clk,
    input reset,
    input d,
    output reg q
);

    always @(posedge clk or posedge reset)
        begin
            if(reset)
                q <= 0;
            else
                q <= d;
            end
endmodule
```

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B. Testbench

```
// Code your testbench here
// or browse Examples
`timescale 1ns/1ps
module tb_async_counter_fault;
    reg clk;
    reg reset;
    reg sa0;
    reg sa1;
    wire [1:0] q;
    wire [1:0] q_fault;
    wire error;
    // Instantiate the DUT
    async_counter_fault uut(
        .clk(clk),
        .reset(reset),
        .sa0(sa0),
        .sa1(sa1),
        .q(q),
        .q_fault(q_fault),
        .error(error)
    );

    // Clock generation
    always #5 clk = ~clk;

    initial
    begin
        clk = 0;
        reset = 1;
        sa0=0;
        sa1=0;

        #10;
        reset = 0; // Apply RESET

        #50; // Normal Operation

        // Inject sa0 fault
        $display("Apply sa0 Fault");
        sa0=1;
        sa1=0;

        #50;
```

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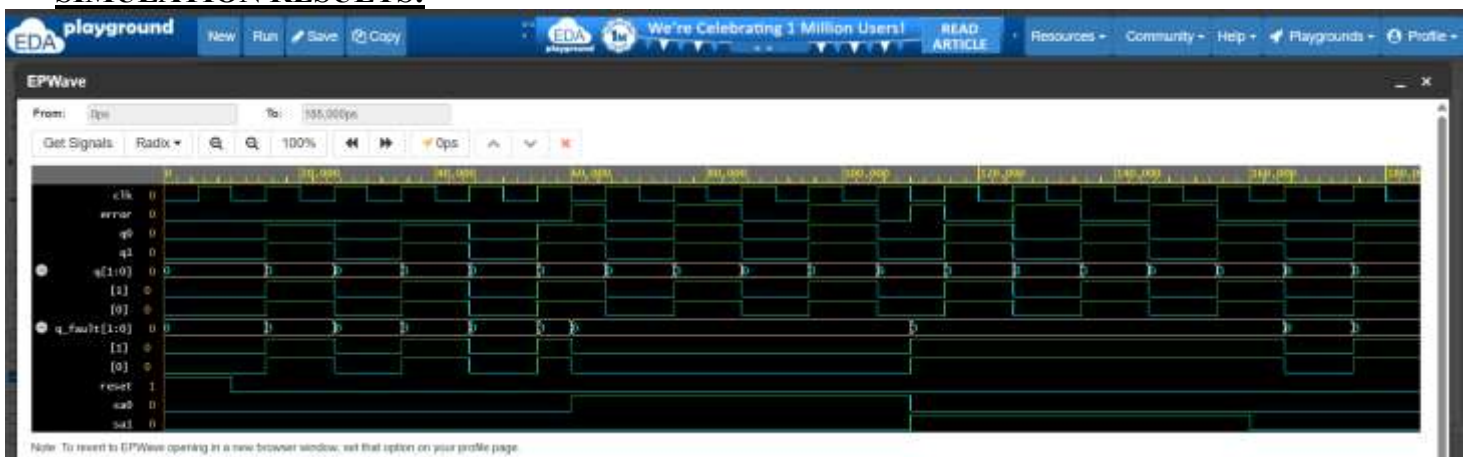
```

sa0=0; // Remove sa0 fault

// Inject sa1 fault
$display("Apply sa1 Fault");
sa1=1;
#50;
sa1=0; // Remove sa1 fault
#25
$finish;
end
// Dump Waveforms
initial begin
  $dumpfile("dump.vcd");
  $dumpvars(0,tb_async_counter_fault);
end
// Monitor values
initial begin
  $monitor("Time=%0t | q = %b | q_fault | sa0 = %b sa1 = %b | error = %b",
$time,q,q_fault,sa0,sa1,error);
end
endmodule

```

SIMULATION RESULTS:



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CONCLUSION:

In this experiment, a 2-bit asynchronous counter using D Flip-Flops was successfully implemented and tested. The counter produced the correct counting sequence, demonstrating the working of ripple counters. Faults such as SA0 and SA1 were introduced, and their effects on the output were observed. This helped in understanding how faults affect circuit behavior and how they can be detected. Overall, the experiment provided knowledge about sequential circuits and fault testing techniques in VLSI systems.

Post Lab Exercise:

To design a 3-bit asynchronous counter using D flipflop.

Design code:

```
// Code your design here
module async_counter_3bit (
    input clk,
    input reset,
    output reg [2:0] Q
);

always @(posedge clk or posedge reset)
begin
    if (reset)
        Q[0] <= 0;
    else
        Q[0] <= ~Q[0];
end

always @(posedge Q[0] or posedge reset)
begin
    if (reset)
        Q[1] <= 0;
    else
        Q[1] <= ~Q[1];
end

always @(posedge Q[1] or posedge reset)
begin
    if (reset)
        Q[2] <= 0;
    else
        Q[2] <= ~Q[2];
end

endmodule
```

Testbench code:

```
// Code your testbench here
// or browse Examples
`timescale 1ns/1ps

module tb_async_counter;
```

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```

reg clk;
reg reset;
wire [2:0] Q;

async_counter_3bit uut (
    .clk(clk),
    .reset(reset),
    .Q(Q)
);

always #5 clk = ~clk;

initial begin
    $dumpfile("counter.vcd");
    $dumpvars(0, tb_async_counter);

    clk = 0;
    reset = 1;

    #10 reset = 0;

    #200 $finish;
end

endmodule

```

Simulation result:

